

FINAL EXPLANATION: BIOLOGY GRADE 11

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY GRADE 11: REPRODUCTION IN ANIMALS (SUB-STRAND 3.1) – FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: How do different animals ensure successful fertilisation and development of offspring, and how can science assist when human reproduction fails? Use what you observed and learned in Lessons 1 to 7 to write your Final Explanation. Answer each prompt in the space below it, using full sentences and the correct scientific words. Refer to the frogs near Kisumu and the Lake Victoria wetlands, the hen, the cow and the IVF stories you met in Lesson 1. Give evidence from your class work, such as the dissection or annotated photographs, the sperm and ovum slides or charts, the hormone graphs, the animations, the case studies and the debate. Your answers will also form part of your portfolio or poster and your position paper on reproductive biotechnology.

Part 1: Frogs, Hens and Cows – Many Ways to One Goal

Frogs near Kisumu release thousands of jelly-covered eggs into water. A hen lays hard-shelled eggs on land. A cow gives birth to one live calf. (a) Explain where fertilisation takes place in the frog, the hen and the cow, and name the type of fertilisation in each. (b) Use the terms oviparous and viviparous to describe how the young of the frog, hen and cow develop. Then explain what ovoviviparous means, using an example from your classification charts. (c) Explain why the frog needs thousands of eggs but the cow produces one calf. Think about the site of fertilisation, losses of eggs and young, and parental care.

(a) In frogs (and fish such as tilapia), the female releases eggs into the water and the male releases sperm over them, so fertilisation happens outside the body. This is external fertilisation. It needs water so that sperm can swim to the eggs. In the hen, fertilisation happens inside the female's body, in the oviduct, before the hard shell forms around the egg. In the cow, fertilisation also happens inside the female's body. Both are internal fertilisation. (b) The frog and the hen are oviparous, because they lay eggs and the young develop in the eggs outside the mother's body. The cow is viviparous: the young develops inside the mother and is born alive. Ovoviviparous animals, such as some snakes, keep the eggs inside the body and the eggs hatch inside the mother. None of the frog, hen or cow is ovoviviparous. (c) In water, fertilisation is by chance. Many eggs are not reached by sperm, and the eggs and tadpoles are unprotected, so many are lost. Producing thousands of eggs makes it likely that some survive to become adults. In the cow, fertilisation happens inside the body, the gametes are protected, the embryo is protected and fed for months, and the mother cares for the calf. Losses are small, so one calf is enough.

Part 2: Built for Reproduction – Structure and Function

(a) From your dissection of a small mammal (or the annotated photographs, models and charts), name the parts of the male system that a sperm passes through, and the main parts of the female system. (b) State what the testes and ovaries do, where fertilisation occurs, and the role of the uterus. (c) Describe the structure of a sperm cell and of an ovum, and explain how each structure helps the cell to do its job (movement, protection or fusion).	(a) In the male, sperm are made in the testes and pass through the epididymis, the sperm duct and the urethra. In the female, the parts are the ovaries, the oviducts, the uterus, the cervix and the vagina. (b) The testes and ovaries produce gametes (sperm and ova). The tubes transport them. Fertilisation occurs in the oviduct. The uterus is muscular and has a good blood supply, so it can hold and support the developing embryo for months. This protected internal development supports the few offspring of mammals such as the cow. (c) The sperm is small and has a tail for swimming. It has many mitochondria that release energy for movement, and an acrosome at the head, which helps the sperm to fuse with the ovum. The ovum is much larger than the sperm. It has food reserves in its cytoplasm to feed the early embryo, and a jelly coat around it that gives protection. Both carry a nucleus with half the genetic material, which join at fertilisation.
--	--

Part 3: Hormonal Conductors – Cycle, Pregnancy and Birth	
(a) Describe how FSH, LH, oestrogen and progesterone change during a 28-day menstrual cycle and what each does. Include what happens near Day 14. (b) Explain how feedback between the pituitary gland and the ovaries helps control the timing of egg release, and compare this with the frog's release of eggs. (c) Explain how hormones and the placenta support the baby if fertilisation occurs, from implantation to birth and lactation. Include hCG, progesterone, oxytocin and prolactin. (d) Suggest what could happen if a hormone is undersecreted or oversecreted.	(a) The pituitary gland releases FSH early in the cycle, which stimulates a follicle in the ovary to develop. The developing follicle makes oestrogen, which rises and builds up the uterine lining. Near Day 14 there is a sharp surge of LH, and ovulation (release of the egg) follows shortly after the LH peak. After ovulation, progesterone rises in the second half of the cycle and maintains the thickened uterine lining. If there is no pregnancy, progesterone falls and menstruation follows. (b) The pituitary hormones (FSH and LH) and the ovarian hormones (oestrogen and progesterone) interact through negative and positive feedback. This feedback times follicle development, the LH surge and ovulation, and prepares the uterine lining. In women, hormone feedback controls when an egg is released, and usually one egg is released each month. Frogs, in contrast, rely on releasing large numbers of eggs into water. (c) If fertilisation occurs, the embryo implants in the uterine wall. It produces hCG, which keeps the corpus luteum making progesterone, so the lining is maintained and there is no menstruation. The placenta forms and the umbilical cord links the baby to it. The placenta exchanges oxygen, nutrients and wastes between mother and baby without mixing their blood, and it also produces hormones. At birth, oxytocin causes strong uterine contractions. After birth, prolactin stimulates milk production (lactation). (d) Too little or too much of a hormone can disrupt follicle development, ovulation or the regularity of the cycle, and can make natural pregnancy difficult. Knowing this helps to clear up myths and to manage menstrual health.

Part 4: When Reproduction Fails – IVF, Surrogacy and Sperm Banking	
(a) Many people think that IVF babies grow wholly in a test	(a) The idea is not correct. In IVF, an egg and sperm are brought together in a dish, so fertilisation happens outside the body, like external fertilisation in frogs. However, the embryo is then transferred to the uterus, where it implants and develops with the help of

tube. Use what you learned to decide whether this is correct and explain what happens in IVF and where the baby develops. (b) For each of the couples in the case studies, say which technology (IVF, surrogacy or sperm banking) could help and why: blocked fallopian tubes (Nakuru), low sperm count (Kisumu), hormonal imbalance (Eldoret), sperm storage before cancer treatment (Nairobi), absent uterus (Mombasa). (c) Describe one ethical issue that must be considered.	hormones and the placenta, as in natural pregnancy. The baby does not grow wholly in a test tube. (b) Blocked tubes (Nakuru): IVF, because fertilisation takes place in a dish, bypassing the blocked tubes, and the embryo is then placed in the uterus. Low sperm count (Kisumu): IVF, because sperm and eggs are brought together in a dish, which gives the sperm a better chance of reaching and fertilising an egg than in the body. Hormonal imbalance (Eldoret): IVF with hormone treatment, because hormones are used to help the eggs mature and be collected, and the embryo is then transferred to the uterus. Cancer treatment (Nairobi): sperm banking, which stores sperm before treatment so the man can use them later. Absent uterus (Mombasa): surrogacy, where another woman carries the embryo, because the embryo needs a uterus to develop. (c) Ethical issues include cost and fair access (many families cannot afford it), consent of everyone involved, and cultural beliefs. Decisions about reproductive health should be shared and informed, by both parents.
---	---

Part 5: Putting It All Together

Write a final answer of about 8 to 10 sentences to the driving question: How do different animals ensure successful fertilisation and development of offspring, and how can science assist when human reproduction fails? Include (1) external and internal fertilisation, with the frog, hen and cow as examples, (2) the role of structure and hormones in humans, (3) the role of biotechnology, and (4) how your thinking has changed since Lesson 1.	Animals ensure successful reproduction with strategies that suit their environment. Frogs and fish in water, such as those near Kisumu and the Lake Victoria wetlands, use external fertilisation and release thousands of eggs, because many are lost to chance. Hens, cows and humans use internal fertilisation, which protects the gametes and the young, so cows and humans need to produce fewer offspring. In humans, the tailed sperm swims and has an acrosome, the larger ovum has food reserves, fertilisation happens in the oviduct, and the uterus supports the embryo. FSH, LH, oestrogen and progesterone work through feedback to time ovulation, and after implantation hCG, progesterone, the placenta, oxytocin and prolactin sustain pregnancy, birth and lactation. When this natural pathway fails, IVF, surrogacy and sperm banking can replace or bypass the failed step, but the embryo still develops in the uterus, so the baby does not grow wholly in a test tube. At first I thought IVF babies grow wholly in a test tube, but now I know that only fertilisation happens in the dish. I can also explain why frogs need thousands of eggs while a cow bears one calf. I understand that these technologies raise questions of cost, consent and culture, and that both parents should share informed decisions.
---	---

RUBRIC

Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Comparing animal strategies	Accurately explains external and internal fertilisation and the oviparous, viviparous and ovoviviparous types with the frog,	Correctly describes the types of fertilisation and development with examples, but the link to the number of	Names some types or examples but with errors, or does not explain why frogs need thousands of eggs and a cow only

	hen and cow, and links the number of offspring to the site of fertilisation, losses and parental care.	offspring is only partly explained.	one calf.
Structure and function	Correctly names the reproductive parts and explains how the sperm, ovum, oviduct and uterus are adapted to their functions, using evidence from the dissection and slides.	Names most parts and describes functions correctly, with limited explanation of how structure suits function.	Names few parts or confuses their functions.
Hormonal control	Clearly explains the roles and timing of FSH, LH, oestrogen and progesterone, feedback loops, and the roles of hCG, the placenta, oxytocin and prolactin, including the effects of imbalance.	Describes most hormones and their roles correctly but does not fully explain feedback or the pregnancy stages.	Mentions hormones without correct roles or sequence.
Biotechnology and ethics	Matches IVF, surrogacy and sperm banking to the specific causes of infertility, corrects the test-tube misconception and discusses ethical issues with reasoning.	Explains the technologies correctly and mentions one ethical issue with little reasoning.	Describes technologies inaccurately or keeps the test-tube misconception.
Communication and use of evidence	Writes a clear, well-organised answer using correct scientific terms and evidence from the lessons, and shows how thinking changed since Lesson 1.	Writes a mostly clear answer with some scientific terms and some evidence.	Writes a brief or unclear answer with few scientific terms and little evidence.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

Licensing & Copyright: © 2026 SeaVuria and ARES. This work is licensed under a [Creative Commons Attribution-NonCommercial 4.0 International License \(CC BY-NC 4.0\)](https://creativecommons.org/licenses/by-nc/4.0/). To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc/4.0/>. You are free to copy, redistribute, and adapt this material in any medium or format, provided appropriate credit is given and the material is not used for commercial purposes. KICD learning outcomes and curriculum designations reproduced here for alignment purposes remain the property of KICD and are not covered by this license. Linked external resources (e.g. CK-12, Khan Academy, PhET) are subject to their own licenses.